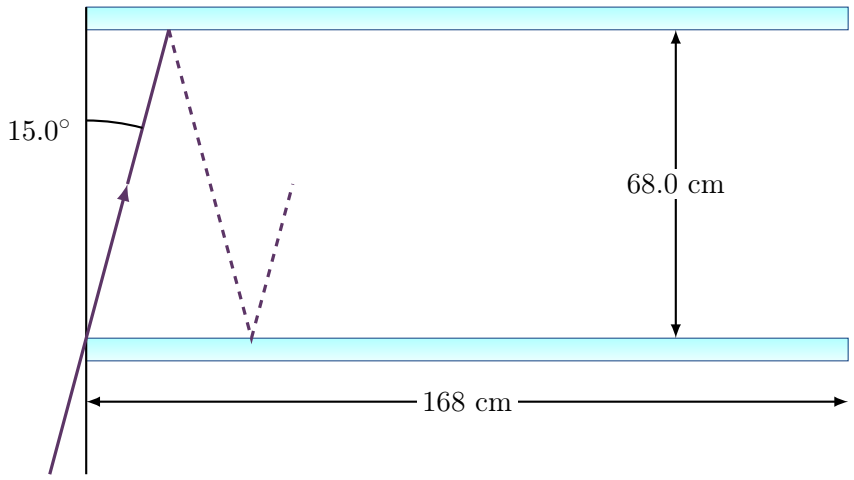
1. A beam of light strikes a mirror at an angle of 23° to the normal. What is the angle between the incident and reflected beams?

2. How many times does the light beam reflect from the top mirror?

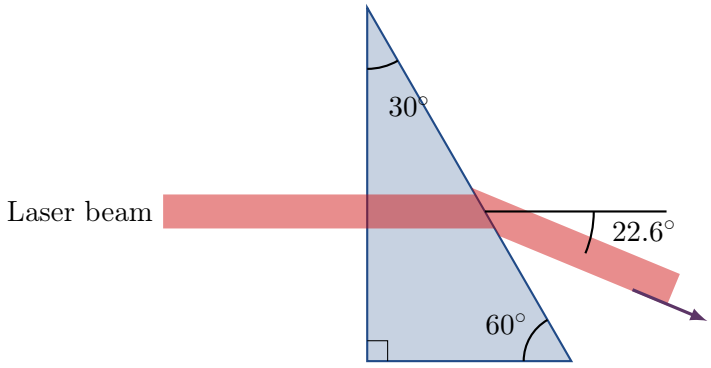


3. A laser beam is aimed at a 1.0-cm thick sheet of glass (n = 1.5) at an angle 30° above the glass.

A. What is the laser beam's direction of travel in the glass?

B. What is its direction in the air on the other side of the piece of glass?

4. The figure shows a laser beam deflected by a 30-60-90 prism. What is the prism's index of refraction?



5. Why do prism cause red and blue light to separate?

6. A sheet of glass has $n_{\text{red}} = 1.52$ and $n_{\text{blue}} = 1.54$. A narrow beam of white light is incident on the glass at 30° from the normal. What is the angular spread of the light inside the glass?

7. What is the critical angle for total internal reflection for a boundary between substance 1 with $n_1 = 1.69$ and $n_2 = 1.44$? In which substance does the TIR occur?

8. Go to `https://demonstrations.wolfram.com/FermatsPrincipleAndSnellsLaw/`. What is the minimum time path of the air/water interface to two significant digits. At which angles does this occur?

9. Get `https://demonstrations.wolfram.com/SnellsLawOfRefractionWaveFronts/`. Set the angle to 1 radian, and see what happens to the wavefronts and ray for medium frequency light through a vacuum-diamond interface.

10. Get `https://demonstrations.wolfram.com/DispersionOfLightThroughAPrism/`. What angle would maximize the dispersion of light through the prism?